
Efficient Ink-Wash Generation with PixArt-Sigma LoRA and 20-to-4-to-2 Step Distillation

Group 19

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Abstract

We present a parameter-efficient framework for adapting and accelerating PixArt- Σ for Chinese ink-wash image generation. A rank-16 LoRA is first trained on a curated 209-image subset to learn domain-specific brushwork, ink diffusion, and negative-space composition. The resulting 20-step Style Teacher is then compressed through latent-trajectory distillation into four- and two-step Students. We evaluate prompt alignment, distributional similarity, and inference latency on 30 held-out prompts with four fixed seeds. The four-step Student preserves the Teacher’s CLIP alignment while providing approximately a sixfold speedup, whereas the two-step Student retains 96.8% of the Teacher’s alignment with an approximately 11.8-fold speedup. These results identify four steps as the quality-oriented operating point and two steps as the latency-oriented alternative for interactive generation.

1 Introduction

Large text-to-image diffusion models provide strong general-purpose generation, but adapting them to a small, coherent artistic domain remains difficult. In our setting, the target is expressive Chinese ink-wash imagery characterized by sparse composition, diffused ink, visible brush variation, and selective colour. A general PixArt- Σ model can respond to explicit style words, yet its outputs do not consistently reproduce the visual statistics of the curated collection. Full fine-tuning is also impractical for our data and compute budget, while the standard 20-step sampler is too slow for interactive use.

We therefore study a combined *adapt-and-accelerate* pipeline. First, Low-Rank Adaptation (LoRA) specializes a frozen PixArt- Σ transformer using paired ink-wash images and captions. Second, the selected 20-step adapter acts as a Teacher whose denoising trajectories supervise progressively faster Students. Our final route reduces inference from 20 transformer evaluations to four and then two. The objective is not only lower latency: the accelerated model must preserve prompt semantics, the learned ink-wash domain, and recognizable object structure. This separation also supports controlled evaluation: Teacher checkpoints are selected on held-out prompts, while Students are compared against the fixed Teacher at matched seeds and conditioning.

2 Preliminaries and Problem Formulation

Latent diffusion. Let E and D be the frozen VAE encoder and decoder, and let $z_0 = E(x)$ denote the clean image latent. For diffusion time t , training constructs

$$z_t = \sqrt{\alpha_t} z_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (1)$$

and a text-conditioned PixArt transformer predicts the denoising target from (z_t, t, c) , where c is the frozen T5 caption embedding. Generation starts from Gaussian noise and repeatedly applies the transformer and scheduler before decoding the final latent with D [1, 3].

Low-rank adaptation. For a frozen linear map $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, LoRA learns

$$W' = W + \frac{\alpha}{r}BA, \quad A \in \mathbb{R}^{r \times d_{\text{in}}}, \quad B \in \mathbb{R}^{d_{\text{out}} \times r}, \quad r \ll \min(d_{\text{in}}, d_{\text{out}}). \quad (2)$$

Only the low-rank matrices are optimized [2]. In our implementation, adapters are attached to attention projections and feed-forward/projection layers; the VAE, T5 encoder, and original PixArt weights remain frozen.

Objective. Given a style dataset $\mathcal{D} = \{(x_i, y_i)\}$, we seek a Teacher adapter ϕ_T and Student adapter ϕ_S that jointly balance style transfer, prompt adherence, structural quality, and computation. Formally, the Student should approximate Teacher transitions on the same latent trajectory while using $K \in \{4, 2\}$ transformer calls rather than 20. This is evaluated with held-out prompts, fixed seeds, CLIP-based prompt alignment, distributional image metrics, and visual inspection; no single metric is treated as a complete measure of image quality.

3 Design

The system has three reusable layers:

$$\text{images/captions} \rightarrow \text{VAE and T5 caches} \rightarrow \text{20-step LoRA Teacher} \rightarrow \text{4-step Student} \rightarrow \text{2-step Student}. \quad (3)$$

The corpus contains 260 image–caption pairs spanning 209 plants, 30 animals, and 21 other subjects; the primary Teacher and distillation route use the 209-image plant subset. We precompute clean 512-pixel VAE latents and length-300 T5 embeddings once, validate their sample fingerprint and ordering, and reuse them across runs. This removes repeated VAE/T5 memory cost and makes Teacher comparisons reproducible on limited hardware.

The Teacher is the frozen PixArt- Σ backbone plus one style LoRA. The Students keep the same backbone and attach trainable low-rank modules to the same target layers, which preserves parameter efficiency and adapter portability. Distillation is performed in latent space: the 4-step Student learns four consecutive five-step Teacher intervals, (0, 5), (5, 10), (10, 15), and (15, 20); the 2-step Student learns two ten-step intervals, (0, 10) and (10, 20). The 4-step adapter initializes the 2-step stage, forming a progressive 20-to-4-to-2 route rather than learning the hardest compression directly [4].

4 Methodology

Teacher training. We encode each caption and image once, then sample a cached clean latent z_0 , time t , and noise ϵ during training. The noisy latent from Eq. (1) is passed through PixArt- Σ with the LoRA adapter, and the adapter is optimized using mean-squared diffusion loss,

$$\mathcal{L}_{\text{teach}} = \mathbb{E}_{(z_0, c), t, \epsilon} \left[\|f_{\theta, \phi_T}(z_t, t, c) - \epsilon\|_2^2 \right]. \quad (4)$$

The primary run uses rank and alpha 16, batch size one, learning rate 10^{-5} , and checkpointing every 1,000 optimizer updates. The Teacher is selected from intermediate checkpoints using held-out prompts and visual inspection, instead of assuming that the final checkpoint generalizes best.

Offline trajectory supervision. With the selected Teacher frozen, we generate deterministic 20-step latent trajectories for 627 prompt variants and two seeds, producing 1,254 trajectories with 21 stored states each. Every record contains its text conditioning, seed, scheduler timesteps, initial noise, and intermediate latents. This cache allows Student training without holding Teacher and Student models in GPU memory simultaneously.

Student loss and progressive compression. Most updates teach a Student to jump between cached phase boundaries. If z_s^T and z_e^T are Teacher states at a phase start and end, the Student prediction $\hat{z}_e^S$ is trained with pseudo-Huber loss

$$\mathcal{L}_{\text{traj}} = \sqrt{\|\hat{z}_e^S - z_e^T\|_2^2 + c^2} - c, \quad c = 10^{-3}. \quad (5)$$

To retain the original diffusion field, 20% of updates instead use the Teacher diffusion target with an MSE anchor loss, while 80% use trajectory targets. The 4-step stage uses a learning rate of 5×10^{-6} . Its adapter then initializes the 2-step stage, which uses 2×10^{-6} . For the second phase of the 2-step Student, half of the starting latents are generated by the current Student rather than copied from the Teacher cache; this on-policy correction reduces train-test mismatch caused by errors accumulated after the first jump.

Teacher evaluation uses classifier-free guidance selected during validation. The distilled Students use guidance scale 1.0 and a single conditional branch, so the reported four-step and two-step models require exactly four and two transformer calls. All experiments use the same cached features, scheduler definition, checkpoint metadata, and fixed evaluation seeds to isolate the effects of LoRA adaptation and step reduction [5].

5 Numerical Experiments

5.1 Protocol and implementation details

We evaluate two questions: whether a LoRA transfers the Chinese ink-wash domain beyond exact training captions, and whether the adapted 20-call model can be compressed to four or two Transformer calls. Experiments use the coherent 209-plant subset of the 260 validated image-caption pairs at 512×512 . The rank-16 Style Teacher uses 20 DPM-Solver++ steps and CFG 1.5. The primary rank-16 Students are the 4,500-update 4-step checkpoint and its 7,000-update 2-step successor; both use CFG 1.0 without an unconditional branch. Full training and cache settings are deferred to the Appendix.

The held-out benchmark contains 30 unseen prompts and four fixed seeds per prompt, or 120 images per model. We report CLIP ViT-B/32 text-image cosine similarity, unbiased squared CMMD against all 209 real plant images, warmed-up median latency, and exact forward-call count. Each seed position forms one 30-prompt replicate; Table 1 reports mean $\pm$ sample SD over four replicates. No additional training was performed for these analyses. CMMD is the unbiased squared MMD between normalized CLIP image features, using a Gaussian RBF kernel of bandwidth 10 and all 209 real plant images as the reference distribution. The same prompts and seed positions are retained for every model, and latency is recorded after warm-up in one software/hardware environment so that only the adapter and call count change.

5.2 Style adaptation and held-out generation

We compared ranks 4, 8, 16, and 32 every 1,000 updates to 10,000 under matched 20-step sampling. The best rewritten-palm result improves CLIP from 0.3399 to 0.3561 (+4.7%), while an unseen misty-mountain prompt improves from 0.3615 to 0.3720 (+2.9%). Because later checkpoints often decline and the best rank is prompt-dependent, held-out selection is more reliable than training longer.

For the preferred unseen fishing-boat prompt at seed 10163 (Figure 1), Teacher B replaces the base model’s photographic appearance with monochrome brushwork and negative space. Both Students preserve the composition, although the 2-step output is softer.

5.3 Few-step quality, efficiency, and seed variance

The 4-step Student preserves 100.65% of Teacher CLIP at $5.99\times$ speedup; the 2-step Student retains 96.84% at $11.78\times$ speedup. Its CMMD lies only 2.00% below the predefined $1.5\times$ -Teacher limit, making four steps the quality-oriented point and two steps the latency-oriented alternative.

The 4-step CLIP advantage (0.0024) is smaller than replicate SD and therefore indicates preservation, not a significant gain. Within-prompt seed variation is larger still (Appendix), so single-seed comparisons are unreliable. Specifically, the root-mean-square within-prompt seed SD is 0.0230, 0.0200, and 0.0213 for Teacher, 4-step, and 2-step respectively. The 2-step CMMD also has the largest replicate SD (1.02×10^{-4}), reinforcing its role as the latency-oriented rather than fidelity-equivalent operating point.

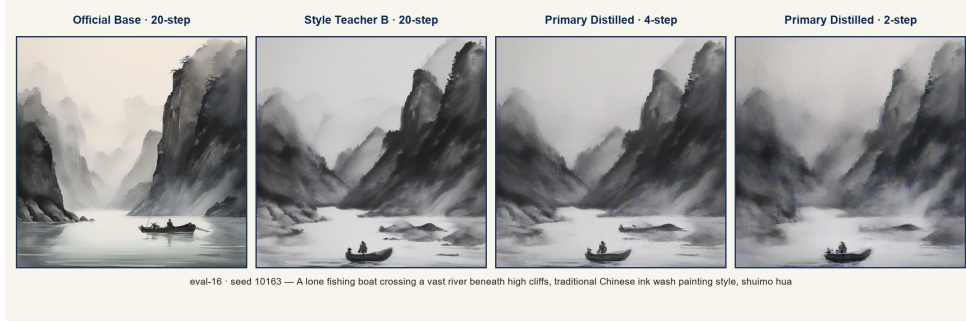


Figure 1: Same unseen prompt and seed for the official 20-step base, 20-step Style Teacher B, primary 4-step student, and primary 2-step student.

Table 1: Held-out mean $\pm$ SD over four 30-prompt seed replicates.

Model	Calls	CLIP $\uparrow$	CMMD $\downarrow$	Time (s) $\downarrow$	Speedup
Teacher B	20	0.3651 ± 0.0040	0.000947 ± 0.000056	2.871 ± 0.006	$1.00\times$
4-step	4	0.3674 ± 0.0049	0.001070 ± 0.000047	0.484 ± 0.009	$5.99\times$
2-step	2	0.3535 ± 0.0025	0.001385 ± 0.000102	0.243 ± 0.010	$11.78\times$

5.4 What happens between the reported endpoints?

Inference-only evaluation on eval-16 (four seeds) decodes every latent state. Figure 2 shows that Teacher CLIP stays near 0.12 through call 14 before reaching 0.346 ± 0.049 at call 20. The 4-step scores are 0.125, 0.122, 0.130, and 0.368 ± 0.020 ; the 2-step scores are 0.122 and 0.366 ± 0.014 . Most visible alignment emerges in the final learned jump. Decoded states appear in the Appendix; early noisy x_t makes per-call CLIP a diagnostic rather than a training objective. The Teacher curve begins its measurable rise only at calls 17–18 (CLIP 0.174 and 0.234), whereas each Student concentrates this transition into its final call. This behavior confirms that few-step inference learns large trajectory jumps rather than merely terminating the original 20-step solver early.

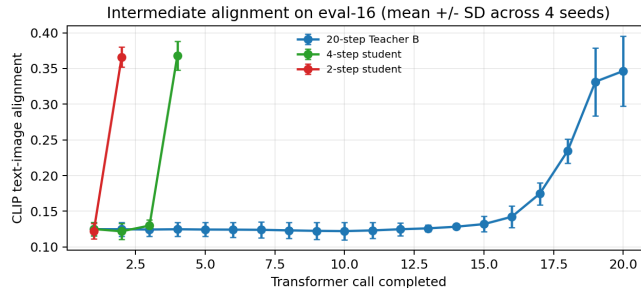


Figure 2: CLIP mean $\pm$ SD over four seeds after every call for the same unseen prompt. The Appendix shows the decoded intermediate states.

6 Discussion

6.1 Answers to core research questions

RQ1: Domain adaptation and generalization under low-rank fine-tuning. Does low-rank adaptation on a small curated dataset transfer Chinese ink-wash style without overfitting?

Rank-16 LoRA (13.8 M parameters, $\sim 0.40\%$ of the backbone) successfully adapts PixArt-Sigma to the Chinese ink-wash aesthetic (*shuimo hua*), capturing monochrome brushwork, wet-ink diffusion on *xuan* paper, and negative space (*liubai*). Generalization on unseen prompts peaks early ($\sim$ step 4,000) before overfitting to training captions. Dataset purity proves decisive: 209 clean plant im-

ages (plant209, CMMD 0.001229) clearly outperform 260 images mixed with photographic noise (n260, CMMD 0.001445), demonstrating that curation quality matters more than scale.

RQ2: Few-step trajectory distillation, guidance elimination, and latency–quality trade-offs. *Can a 20-step teacher be compressed into 4-step and 2-step students while preserving quality, and what governs the step-count trade-off?*

Yes. Trajectory distillation (80% trajectory loss + 20% diffusion anchor) successfully compresses the 20-step teacher into few-step regimes. Training students at CFG = 1.0 eliminates the negative guidance pass, halving the required forward passes per step. The 4-step student achieves a $5.99\times$ speedup (0.480 s vs. 2.871 s) with complete quality retention (100.65% CLIP alignment at 0.3674 ± 0.0049 and low CMMD 0.001070 ± 0.000047). Distilling to 8 or 16 steps is unnecessary because visual fidelity already saturates at 4 steps, adding a $2\times$ – $4\times$ latency penalty without measurable gain. Conversely, pushing distillation to 2 steps unlocks the real-time interactive regime (0.243 s, $11.78\times$ speedup) with 96.84% CLIP retention (0.3535 ± 0.0025) and acceptable style distance (CMMD 0.001385 ± 0.000102 , within the $1.5\times$ teacher limit), establishing 4 steps as the fidelity-oriented benchmark and 2 steps as the speed-oriented deployment point.

RQ3: Intermediate-step dynamics, sampling mechanics, and few-step jump distillation. *What occurs at each step during 20-step generation, how does training differ from sampling, and how do few-step students compress these dynamics?*

Sampling steps and **training updates** operate under distinct mechanics. During 20-step Teacher sampling, solving the reverse ODE at CFG = 1.5 requires 40 forward passes. Decoding intermediate latents x_t shows that visual emergence is non-linear: steps 1–14 ($t \in [999, 300]$) remain off-manifold noise (CLIP ≈ 0.12), coarse composition emerges at steps 15–17, and fine brushwork with negative space resolves only in the final two steps ($t \leq 100$, CLIP ≥ 0.35). Crucially, Teacher **training** samples a single random t per update rather than rolling out 20 steps. **Student distillation** bridges this gap: using offline cached trajectories, 4-step and 2-step students learn direct macroscopic jumps (spanning 5 and 10 teacher steps per call) onto the clean manifold at CFG = 1.0, bypassing numerical ODE solver failure.

6.2 Variance dynamics and evaluation reliability

Across 30 test prompts, seed variation (RMS SD ≈ 0.020 – 0.023) exceeds model-level differences. Evaluating 4 fixed seeds across held-out prompts (120 images per model) prevents single-seed bias and confirms that the 4-step student reliably matches teacher performance.

7 Conclusions

7.1 Summary of Contributions and Key Insights

This work demonstrates an efficient adapt-and-accelerate framework for specialized artistic diffusion synthesis. First, a lightweight rank-16 LoRA adapter (13.76 M parameters, $\sim 0.40\%$ of the DiT backbone) trained on a curated 209-image plant corpus reliably captures wet-ink diffusion on *xuan* paper and negative space (*liubai*), showing that dataset purity and early stopping ($\sim$ step 4,000) outweigh dataset scale. Second, progressive trajectory distillation ($20 \rightarrow 4 \rightarrow 2$ steps) combined with classifier-free guidance elimination (CFG = 1.0) and 50% on-policy roll-in establishes an optimal Pareto frontier: the 4-step student provides the quality-oriented regime (0.480 s, $5.99\times$ speedup, 100.65% CLIP retention), while the 2-step student unlocks real-time interactive generation (0.243 s, $11.78\times$ speedup, 96.84% CLIP retention). Finally, intermediate latent analysis reveals that image emergence in diffusion ODEs is non-linear, and distilled students successfully bypass numerical solver failure at ≤ 4 steps by executing direct learned macroscopic jumps.

7.2 Limitations and Future Work

While effective, our evaluation was conducted at 512×512 on floral/landscape compositions, and the 2-step student exhibits slight softening of dry-brush micro-textures relative to the teacher. Future research will explore single-step consistency distillation and rectified flow for sub-100 ms generation, adversarial objectives to sharpen ultra-few-step brushwork, and domain-specific quantitative metrics measuring ink bleeding gradients and negative space ratios.

References

- [1] J. Chen et al. PixArt- Σ : Weak-to-Strong Training of Diffusion Transformer for 4K Text-to-Image Generation. In *European Conference on Computer Vision*, 2024.
- [2] E. J. Hu et al. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*, 2022.
- [3] J. Song, C. Meng, and S. Ermon. Denoising Diffusion Implicit Models. In *International Conference on Learning Representations*, 2021.
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- [5] Group 19. Efficient PixArt-Sigma LoRA. <https://github.com/jackyckp/efficient-pixart-sigma-lora/tree/ZHS>, 2026.

A Reproducibility and Additional Results

A.1 Exact primary-model configuration

Table 2 records implementation details omitted from the main numerical section for space. Cached VAE latents are FP16 tensors of shape $4 \times 64 \times 64$ and T5-XXL prompt embeddings have shape 300×4096 . The prompt bank contains three variants of each of 209 plant captions and two deterministic seeds per prompt, producing 1,254 teacher trajectories.

Table 2: Primary distillation configuration.

Component	Optimization	Inference / phase mapping
Teacher B	Rank/alpha 16/16 style LoRA	20-step DPM-Solver++, CFG 1.5
4-step student	LR 5×10^{-6} ; selected at 4,500/6,000 updates	CFG 1.0; trajectory indices $0 \rightarrow 5 \rightarrow 10 \rightarrow 15 \rightarrow 20$
2-step student	LR 2×10^{-6} ; 7,000 updates; 50% on-policy roll-in	CFG 1.0; trajectory indices $0 \rightarrow 10 \rightarrow 20$
Both students	Rank/alpha 16/16; 80% pseudo-Huber trajectory loss; 20% clean-latent anchor	FP16 base, FP32 LoRA; no unconditional CFG branch

A.2 Seed-resolved metrics

Each column s_1-s_4 in Table 3 is the mean CLIP score for the same 30 unseen prompts under one seed position. The final column instead computes the root-mean-square of the within-prompt SD across four seeds. The 4-step aggregate gain is smaller than replicate variation, while all three models show much larger example-level seed sensitivity.

Table 3: Seed-resolved CLIP results underlying the mean $\pm$ SD in Table 1.

Model	s_1	s_2	s_3	s_4	Replicate variance	Within-prompt RMS SD
Teacher B	0.3634	0.3653	0.3610	0.3705	1.63×10^{-5}	0.0230
4-step	0.3685	0.3653	0.3623	0.3736	2.36×10^{-5}	0.0200
2-step	0.3539	0.3501	0.3543	0.3559	6.01×10^{-6}	0.0213

A.3 Decoded intermediate states

Figure 3 complements the per-call curve in Figure 2 with the actual decoded states for seed 10163. The teacher schedule is sampled at calls 1, 5, 10, 15, 18, and 20; every student call is shown. The intermediate images and CLIP features were cached during a single inference-only pass and reused by subsequent report evaluation.

Denoising intermediates: eval-16, seed 10163

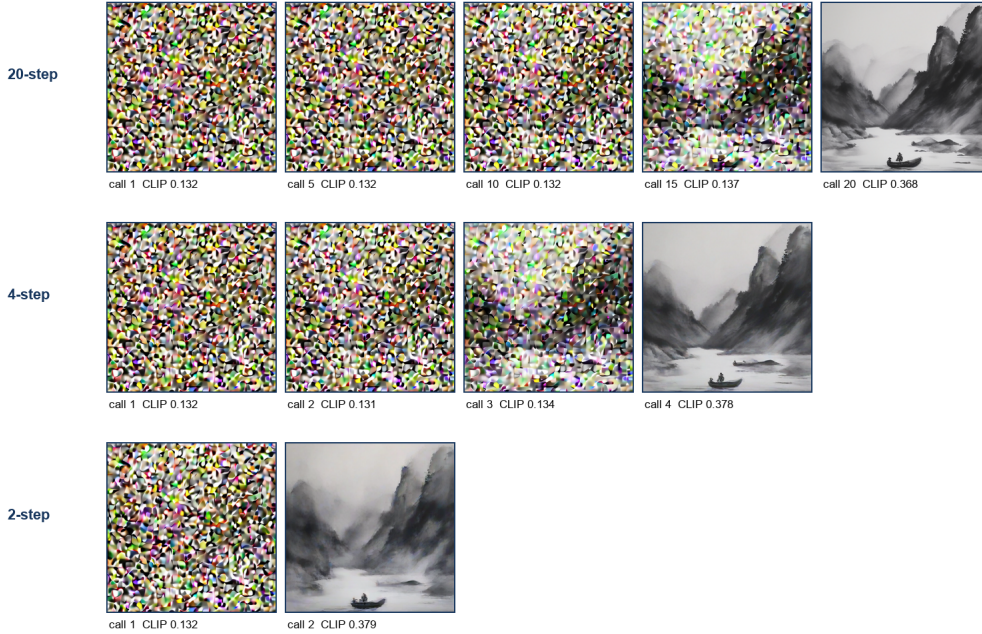


Figure 3: Decoded denoising states for the preferred held-out fishing-boat prompt at seed 10163. Composition becomes visually measurable mainly in the last teacher or student transition.

For reference, the teacher CLIP mean stays near 0.12 through call 14, then rises to 0.174 (call 17), 0.234 (call 18), and 0.346 ± 0.049 (call 20). The 4-step sequence is 0.125, 0.122, 0.130, and 0.368 ± 0.020 ; the 2-step sequence is 0.122 and 0.366 ± 0.014 . These values diagnose visible emergence only: intermediate x_t is intentionally noisy and need not lie on the natural-image manifold. Per-call images, per-seed scores, and feature caches are retained in the evaluation outputs for reproducibility.

A.4 Architecture and cached representations

Component	Setting
Backbone	PixArt- Σ XL 2, 512-pixel variant
Full corpus	260 image-caption pairs
Primary style subset	209 plant image-caption pairs
Other corpus categories	30 animal and 21 other pairs
Cached image features	FP16 VAE latents, $[N, 4, 64, 64]$
Cached text features	FP16 T5 embeddings, $[N, 300, 4096]$
Teacher adapter	rank 16, alpha 16, learning rate 10^{-5}
4-step Student	four five-step phases, learning rate 5×10^{-6}
2-step Student	two ten-step phases, learning rate 2×10^{-6}
Trajectory cache	627 prompts, two seeds, 1,254 trajectories
Student task sampling	80% trajectory, 20% diffusion anchor

Table 4: Full implementation settings used by the Teacher-Student pipeline.

Text conditioning and cross-attention. The VAE and T5 encoder are frozen and evaluated before LoRA optimization. The VAE converts each training image into a clean latent, while T5 converts its caption into a sequence of contextual embeddings. During Transformer training, the noisy image-latent tokens attend to these cached text features through the model’s cross-attention blocks. The method therefore trains neither the VAE nor the text embeddings; it updates only the attached LoRA matrices while the frozen text features condition the denoising prediction.

LoRA target modules. The implementation is not restricted to attention weights. LoRA modules are attached to query, key, value, and attention-output projections, together with feed-forward and additional projection layers. Representative targets include `ff.net.0.proj`, `ff.net.2.proj_in`, `proj_out`, `proj`, `linear`, `linear_1`, and `linear_2`. The original PixArt parameters remain frozen.

A.5 Cache validation

All images, captions, latents, and text embeddings are linked by a deterministic manifest. Before training, the loader checks sample identifiers, ordering, tensor shapes, data types, finite values, and the dataset fingerprint. This prevents a valid-looking latent from being paired silently with the wrong text embedding. The validated cache can then be reused across LoRA ranks, learning rates, checkpoints, and Student stages.

B Extended Distillation Procedure

B.1 Teacher trajectory construction

The selected 20-step Teacher is loaded once to generate deterministic latent trajectories. Each cached record contains 21 latent states, its prompt index, seed, scheduler configuration, Teacher-adapter hash, and prompt-bank fingerprint. After the cache is verified, the Teacher is removed from GPU memory and Student optimization uses only cached trajectory states and text conditioning.

Stage	Teacher-state intervals	Student calls
4-step Student	(0, 5), (5, 10), (10, 15), (15, 20)	4
2-step Student	(0, 10), (10, 20)	2

Table 5: Cached Teacher intervals learned by each Student.

B.2 Trajectory and anchor objectives

For a predicted phase endpoint $\hat{z}_e^S$ and cached Teacher endpoint z_e^T , trajectory updates use the element-wise pseudo-Huber loss

$$\mathcal{L}_{\text{traj}} = \text{mean} \left(\sqrt{(\hat{z}_e^S - z_e^T)^2 + c^2} - c \right), \quad c = 10^{-3}. \quad (6)$$

With probability 0.8, an optimizer update uses $\mathcal{L}_{\text{traj}}$; with probability 0.2, it instead applies the standard diffusion noise-prediction MSE to a real clean latent. Thus, 80/20 is stochastic task sampling rather than two losses evaluated in every update.

For the second phase of the 2-step Student, half of the starting states are generated by the current Student’s first jump and detached before the second jump. This on-policy correction exposes the model to its own first-step error and reduces the train–test mismatch that would arise if every second-phase input were copied from a perfect Teacher trajectory.

B.3 Training sequence

1. Encode images with the frozen VAE and captions with the frozen T5 encoder; validate identifiers, shapes, ordering, and hashes.
2. Attach rank- r LoRA modules to the frozen PixArt Transformer and train the 20-step style Teacher with diffusion MSE.
3. Select one Teacher checkpoint using held-out prompts rather than assuming that the final training checkpoint is optimal.
4. Cache deterministic 20-step Teacher trajectories for the expanded prompt bank and unload the Teacher.
5. Initialize the 4-step Student from the selected Teacher LoRA and train it on the four intervals in Table 5.

6. Select an accepted 4-step checkpoint, initialize the 2-step Student from it, and train with two intervals and on-policy correction.
7. Evaluate fixed checkpoints using matched prompts, seeds, schedulers, and exact Transformer-call counts.

C Reproducibility Information

Each reported run should record the following fields:

- base-model identifier and revision;
- dataset, latent-cache, text-cache, and prompt-bank fingerprints;
- LoRA target modules, rank, alpha, learning rate, and checkpoint;
- scheduler class, complete timestep sequence, inference steps, and guidance scale;
- training and evaluation seeds, prompt identifiers, and sample count;
- Teacher and Student adapter hashes; and
- measured latency, hardware, precision, memory mode, and exact number of Transformer forward calls.